

Website Security Scanner

Detailed Weekly Progress Report

Week 1: Project Planning, UI Design & Authentication

Monday

Tasks Completed:

- Analyzed project requirements and objectives.
- Researched website security concepts and OWASP vulnerabilities.
- Finalized project scope and workflow.
- Selected HTML, CSS, JavaScript and FastAPI as technology stack.

Deliverables:

- Requirement analysis completed.
- Technology stack finalized.

Tuesday

Tasks Completed:

- Designed project architecture.
- Created project folder structure.
- Prepared frontend and backend directories.
- Configured development environment.

Deliverables:

- Project structure initialized.

Wednesday

Tasks Completed:

- Developed responsive landing page.
- Created navigation bar and hero section.
- Added Features section.

Deliverables:

- Homepage UI completed.

Thursday

Tasks Completed:

- Designed Pricing, FAQ and Footer sections.
- Improved responsive layout.
- Enhanced UI styling.

Deliverables:

- Landing page finalized.

Friday

Tasks Completed:

- Developed Login and Signup pages.
- Created authentication forms.
- Added client-side validation.

Deliverables:

- Authentication UI completed.

Saturday

Tasks Completed:

- Integrated authentication backend.
- Implemented session handling.
- Performed login/signup testing.
- Fixed UI bugs.

Deliverables:

- Authentication module completed.

Week 2: Backend Development & Security Scanning

Monday

Tasks Completed:

- Configured FastAPI backend.
- Created API routes.
- Connected frontend with backend.

Deliverables:

- Backend initialized.

Tuesday

Tasks Completed:

- Developed SSL Certificate Scanner.
- Extracted certificate information.
- Handled SSL errors.

Deliverables:

- SSL module completed.

Wednesday

Tasks Completed:

- Implemented Security Headers Scanner.
- Checked HTTP response headers.
- Generated security recommendations.

Deliverables:

- Headers module completed.

Thursday

Tasks Completed:

- Built DNS lookup functionality.
- Collected domain information.
- Validated domain responses.

Deliverables:

- DNS scanner completed.

Friday

Tasks Completed:

- Developed Port Scanner using Python sockets.
- Scanned common ports.
- Improved timeout handling.

Deliverables:

- Port scanner completed.

Saturday

Tasks Completed:

- Tested APIs using Postman.
- Validated error handling.
- Optimized API responses.

Deliverables:

- Core backend verified.

Week 3: SEO, Performance & Scoring System

Monday

Tasks Completed:

- Created SEO analysis module.
- Implemented title tag analysis.
- Checked meta descriptions.

Deliverables:

- SEO scanner started.

Tuesday

Tasks Completed:

- Added H1 and image ALT analysis.
- Detected missing SEO elements.
- Generated SEO recommendations.

Deliverables:

- SEO module completed.

Wednesday

Tasks Completed:

- Integrated Google PageSpeed Insights API.
- Collected performance metrics.
- Displayed optimization suggestions.

Deliverables:

- Performance scanner integrated.

Thursday

Tasks Completed:

- Developed technology detection module.
- Identified frameworks, libraries and server technologies.

Deliverables:

- Technology scanner completed.

Friday

Tasks Completed:

- Created security scoring algorithm.
- Calculated overall security score.
- Added risk classification.

Deliverables:

- Scoring engine completed.

Saturday

Tasks Completed:

- Integrated recommendation engine.
- Improved result accuracy.
- Performed module testing.

Deliverables:

- Analysis modules finalized.

Week 4: Dashboard Integration & Report Generation

Monday

Tasks Completed:

- Integrated frontend with backend APIs.
- Displayed scan results dynamically.

Deliverables:

- API integration completed.

Tuesday

Tasks Completed:

- Developed dashboard cards for SSL, SEO, Ports and Headers.
- Improved data visualization.

Deliverables:

- Dashboard completed.

Wednesday

Tasks Completed:

- Implemented report download feature.
- Formatted scan reports.

Deliverables:

- Report generation completed.

Thursday

Tasks Completed:

- Added loading indicators and better error messages.
- Enhanced user experience.

Deliverables:

- UI improvements completed.

Friday

Tasks Completed:

- Optimized backend performance.

- Reduced API response time.
- Fixed identified bugs.

Deliverables:

- Performance optimized.

Saturday

Tasks Completed:

- Conducted integration testing.
- Verified all modules together.

Deliverables:

- System ready for deployment.

Week 5: Deployment, Testing & Documentation

Monday

Tasks Completed:

- Performed complete system testing.
- Verified scanner accuracy.
- Resolved remaining issues.

Deliverables:

- Testing completed.

Tuesday

Tasks Completed:

- Conducted cross-browser compatibility testing.
- Validated responsive design.

Deliverables:

- Frontend verified.

Wednesday

Tasks Completed:

- Deployed backend on Render.
- Configured environment variables.

Deliverables:

- Backend deployed.

Thursday

Tasks Completed:

- Deployed frontend on Netlify.
- Connected live backend APIs.

Deliverables:

- Application deployed.

Friday

Tasks Completed:

- Prepared project documentation.
- Updated README and usage guide.
- Captured project screenshots.

Deliverables:

- Documentation completed.

Saturday

Tasks Completed:

- Performed final validation.
- Reviewed project deliverables.
- Prepared final submission.

Deliverables:

- Project successfully completed.